



MOLECULE GENERATION CHALLENGE

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GOAL: GENERATING REALISTIC MOLECULES FROM SMILES

THE PROBLEM

What is the goal?

- generate realistic molecules from SMILES training data

TODO

- Model must learn the distribution of human-designed molecules

Constraint 1

- Gen. molecules should be chemically valid

Constraint 2

- Gen. molecules shall be new and not copied from train set

Main challenge?

- Balance realism and novelty

BACKGROUND INFORMATION

- Molecules are represented as SMILES strings
- SMILES encode atoms, bonds, branches, and rings as text
- Molecule generation is treated like a language modeling task
- model learns token patterns from existing molecular structures
- RDKit is used to parse, validate, and canonicalize molecules

DATASET AND CHALLENGE SETUP

- Training data: 1,272,851 human-designed SMILES
- Submission file: 10,000 generated SMILES (own decision)
- Each line contains one molecule
- Evaluation uses the provided FCD metric script
- Positive grade constraints:
 - FCD score < 8.0
 - uniqueness, novelty & validity > 90%

DATA PREPROCESSING AND REPRESENTATION

- SMILES parsed with RDKit
- Invalid molecules were removed
- Molecules were canonicalized for reliable comparison
- SMILES were tokenized into chemically meaningful tokens
- A vocabulary of 100 SMILES tokens was created
 - Examples: Cl, Br, [nH], =, (, 1, etc.

MODELING STRATEGY

- A GRU-based language model was trained on SMILES tokens
- The model predicts the next token given the previous tokens
- Training objective: next-token prediction
- Generation starts with a beginning token and ends with an end token
- Sampling uses temperature and top-k filtering
- Lower temperature gives more conservative molecules
- Higher temperature increases diversity but may reduce realism

QUALITY CONTROL OF MOLECULES

- Generated candidates were checked using RDKit
- Invalid SMILES were discarded
- Molecules already present in the training set were removed
- Duplicate generated molecules were removed
- Atom-count filtering was used to avoid unrealistic extremes
- Final output was written to submission.txt

RESULTS

- Device used: CUDA-enabled GPU (2080 Super)
- Training set size after remained the same (no augmentation used)
- Training loss decreased over 5 epochs
 - Epoch 1: 0.7897
 - Epoch 5: 6490
- Generated 10,000 molecules
- Uniqueness, Validity & Novelty all at 100%
- FCD: 0.99

KEY FINDINGS & INTERPRETATION

- The model successfully learned valid SMILES syntax
- RDKit filtering ensured high validity
- Canonical comparison prevented duplicates and training-set copies
- The achieved FCD of 0.99 corresponds to grade range 1
- Sampling parameters strongly influence the FCD score
- Conservative sampling can improve realism but may reduce diversity

LIMITATIONS & FUTURE WORK

- Limitations:

- Model is sequence-based and does not explicitly understand molecular graphs
- SMILES syntax validity does not guarantee chemical usefulness
- FCD optimization depends strongly on sampling parameters
- More training could improve distribution matching

- Future Work:

- tune temperature and top-k systematically
- add molecular property filters
- use graph-based generative models
- train larger or transformer-based models